3.2.6 Cost of Gram-Schmidt algorithms ¶

(No video for this unit.)

Homework 3.2.6.1. Analyze the cost of the CGS algorithm in Figure 3.2.4.2 (left) assuming that $A \in \mathbb{C}^{m \times n}$.

▼ Solution

During the kth iteration ($0 \leq k < n$), A_0 has k columns and A_2 has $n-k-1$ columns. In each iteration

Operation	Approximate cost (in flops)
$r_{01} := A_0^H a_1$	$2mk$
$a_1 := a_1 - A_0 r_{01}$	$2mk$
$\rho_{11} := \\|a_1\\|_2$	$2m$
$a_1 := a_1 / \rho_{11}$	m

Thus, the total cost is (approximately)

$$
\begin{array}{l}
\sum_{k=0}^{n-1} [2mk + 2mk + 2m + m] \\
\quad = \\
\sum_{k=0}^{n-1} [3m + 4mk] \\
\quad = \\
3mn + 4m \sum_{k=0}^{n-1} k \\
\quad \approx \quad < \sum_{k=0}^{n-1} k = n(n-1)/2 \approx n^2/2 > \\
3mn + 4m \frac{n^2}{2} \\
\quad = \\
3mn + 2mn^2 \\
\quad \approx \quad < 3mn \text{ is of lower order } > \\
2mn^2
\end{array}
$$

Homework 3.2.6.2. Analyze the cost of the MGS algorithm in Figure 3.2.4.2 (right) assuming that $A \in \mathbb{C}^{m \times n}$.

▼ Solution

During the kth iteration ($0 \leq k < n$), A_0 has k columns. and A_2 has $n-k-1$ columns. In each iteration

Operation	Approximate cost (in flops)
$\rho_{11} := \|a_1\|_2$	$2m$
$a_1 := a_1/\rho_{11}$	m
$r_{12}^T := a_1^H A_2$	$2m(n-k-1)$
$A_2 := A_2 - a_1 r_{12}^T$	$2m(n-k-1)$

Thus, the total cost is (approximately)

$$
\begin{aligned}
& \textstyle\sum_{k=0}^{n-1} [2m(n-k-1) + 2m(n-k-1) + 2m + m] \\
& \quad = \\
& \textstyle\sum_{k=0}^{n-1} [3m + 4m(n-k-1)] \\
& \quad = \\
& 3mn + 4m \textstyle\sum_{k=0}^{n-1} (n-k-1) \\
& \quad = \quad < \text{Substitute } j = (n-k-1) > \\
& 3mn + 4m \textstyle\sum_{j=0}^{n-1} j \\
& \quad \approx \quad < \textstyle\sum_{j=0}^{n-1} j = n(n-1)/2 \approx n^2/2 > \\
& 3mn + 4m \frac{n^2}{2} \\
& \quad = \\
& 3mn + 2mn^2 \\
& \quad \approx \quad < 3mn \text{ is of lower order} > \\
& 2mn^2
\end{aligned}
$$

Homework 3.2.6.3. Which algorithm requires more flops?

▼ Solution

They require the approximately same number of flops.

A more careful analysis shows that, in exact arithmetic, they perform exactly the same computations, but in a different order. Hence the number of flops is exactly the same.